



The Bombay Salesian Society's
DON BOSCO INSTITUTE OF TECHNOLOGY

Premier Automobiles Road, Kurla West, Mumbai - 400070

Department of Electronics & Telecommunication Engineering

Report on Industrial Visit

Title: Industrial Visit to satellite earth station Yeur

Date: March 28, 2025

Time: 8.30am - 5pm

Venue: BSNL Satellite earth station

Target Audience: B.E Sem VIII Electronics & Telecommunication Engineering

No. of Participants Present: 30

Resource Person: Divisional Engineer, BSNL Satellite earth station

Organizing Department / Committee / Authority: Electronics & Telecommunication

Faculty Coordinator: Aparna M. Telgote

Objectives:

- ❖ To bridge the gap between industry and academia
- ❖ To make students aware of career opportunities in the field of Telecommunication domain

Detailed Report:



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On March 28, 2025, a cohort of 27 students pursuing Bachelors in Engineering in the domain of Electronics and Telecommunications, accompanied by 3 educators **Prof. Aparna Telgote, Prof. Poonam Chakraborty and Mr. Deepak Pandav** from Don Bosco Institute of Technology, Kurla, embarked on an industrial visit expedition to the BSNL Satellite Earth Station nestled amidst the serene environs of Yeur Hills, Thane. This immersive half-day session not only broadened our horizons but also deepened our understanding of satellite communication systems, antennas, and associated equipment, unveiling the intricacies of telecommunications technology.

The session commenced with a session, informing us of the operational intricacies of the antennas, the equipment, and systems set-up within the satellite earth station. Through insights interwoven with real-world examples, we traversed the concepts of signal modulation, transmission protocols, and network management systems (NMS), gaining invaluable information, such as the operational dynamics of satellite communication networks.

Venturing to the height of the facility, we were enthralled by the sight of towering antennas, meticulously aligned to establish robust communication links spanning vast distances. These antennas, boasting impressive technical specifications such as diameter, gain, and beamwidth, serve as the linchpin of satellite communication infrastructure, enabling seamless connectivity across continents.

One notable antenna we encountered was a parabolic reflector with a diameter of 9 meters and a gain of 40 dB, operating within the Ku-band frequency range (12-18 GHz). This antenna, equipped with sophisticated tracking mechanisms, enables precise alignment with geostationary satellites, facilitating high-speed data transmission over long distances.

Furthermore, our exploration delved into the realm of frequency spectrum allocation, where we gained statistical insights into the utilization of various frequency bands. The Ku-band, characterized by frequencies ranging from 12 to 18 GHz, finds widespread application in satellite communication systems, offering high data throughput and efficient spectrum utilization. In contrast, the Ka-band, operating within the frequency range of 26.5 to 40 GHz, is prized for its immense bandwidth capacity, catering to emerging applications such as satellite internet and broadband services.

A focal point of our visit was the symbiotic relationship between BSNL and prominent satellite providers like IPSTAR and Thaicom. IPSTAR, renowned for its state-of-the-art routers and network management solutions, augments the operational efficiency of satellite communication systems, ensuring seamless data transmission and network optimization. Thaicom, a leading satellite operator in the Asia-Pacific region, boasts a constellation of geostationary satellites, strategically positioned to deliver a wide array of telecommunication services, including broadcasting, broadband internet, and mobile communication.

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Enclosed : Attendance

Office of the Divisional Engineer,
Satellite Earth Station, Yeur
Core Network Transmission West, Mumbai
J.K.Gram, Yeur, Thane-400 606.
☎ : 8275006548, 022-25888091



भारत संचार निगम लिमिटेड
(भारत सरकार का उपक्रम)
BHARAT SANCHAR NIGAM LIMITED
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No.DE SM/Yeur/T-12/Trainee/2024-25/10

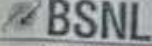
Dated at Yeur, the 28.03.2025

To,
The Principal,
Don Bosco Inst. Of Technology,
Kurla(W), Mumbai-400070

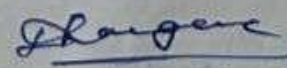
This is to inform that the following students Don Bosco Institute of Technology, Premier Automobiles Road, Kurla(W)-400070 of have successfully attended Industrial Visit scheduled on 28.03.2025 at Satellite Earth Station, BSNL-Yeur.

Sr. No.	Students Name
1	Sumith Kumar
2	Nikhilesh Tumma
3	Mayur Sonar
4	Diya Rai
5	Yash Garje
6	Akshay Talckar
7	Ananya Boliyar
8	Kshitija Zodge
9	Preetam Ghadge
10	Mrunal suplekar
11	Kirti Teli
12	Aditya Mohan Murbade
13	Ann Vincent Chittilappilly Ambili
14	Atharv Parrey
15	Shivraj Shetty
16	Dibyarupa Pradhan
17	Rohan Kadam
18	Kartik Kunder
19	Rohit Mali
20	Suhani Sawant
21	Amey Bhombade
22	Pooja Kadam
23	Kiran Rupanwar
24	Anup Chiplunkar
25	Ansari Mohd Raza
26	Palak Tiwari
27	Sakshi Shidruk

Faculty: (1) Ms. Aparna M. Telgote, (2) Ms. Poonam Chakraborty, (3) Mr. Deepak Pandav
Place: Yeur, Thane
Date: 28/03/2025

 **BSNL**
VSAT




Divisional Engineer (SM)
Bharat Sanchar Nigam Ltd.
Satellite Earth Station, CN-TX West BSNL,
Yeur, Thane - 400 606.